Team Number: 89

M2 — Automatic Category Suggestions for Wikimedia Commons

Problem Statement 3 · Progress report

1. Current approach and changes since M1

The M1 architecture is fully implemented and unchanged: a frozen CLIP ViT-B/32 encoder; four scorers (k-NN voting over categorised images, zero-shot category-name prototypes, a text-initialised multi-label head trained with asymmetric loss, and a metadata scorer using title/description embedding, word overlap and year agreement); logistic-regression fusion over the union of candidates; hierarchy post-processing; live index updates from editor feedback and new categories. Working end to end: Commons crawler, training, evaluation harness, command-line predictor, and a web demo with evidence images and accept/reject feedback. Changes since M1:

- **Hierarchy strength tuned on validation** instead of fixed by hand (1.0 = suppression off), after Run 1 showed a fixed penalty lowered strict F1@5.
- **Larger crawl** (more files per category and subcategory) and a fix for a crawler edge case in the Commons API response.
- **Portable search:** exact NumPy search on macOS, where FAISS and PyTorch can deadlock (both bundle OpenMP); FAISS (flat/HNSW/IVF-PQ) elsewhere.

2. Experiments and results

Setup. Commons files crawled through the MediaWiki API from 107 seed categories (objects, scenes, events, abstract concepts; 12 countries in Asia, Africa and Latin America) and their subcategories. Labels are editor-assigned categories after removing hidden and maintenance categories. Temporal split by upload time; no near-duplicates found between splits (perceptual hash + embedding cosine ≥ 0.95). Run 1: 1,279 files (898/191/190). **Run 2: 3,416 files (2,400 train / 506 val / 510 test)**, vocabulary 11,821 categories, 415 with ≥ 3 training images in the supervised head; k-NN tuned on validation to K = 100, τ = 0.02.

Table 1. Scaling the data (test set, image + metadata).

metric	Run 1	Run 2
F1@5	0.119	**0.160**
P@1	0.284	**0.339**
R@10	0.238	**0.310**
hierarchy-aware P@1	0.384	**0.482**
F1@5, image only	0.089	**0.112**
label coverage	0.335	0.377
mAP (macro)	0.174	0.142
test categories	288	622

Table 2. Run 2, fusion + hierarchy (sample-wise).

input	k	P@k	R@k	F1@k	hier. P
image + metadata	1	0.339	0.159	0.199	0.482
image + metadata	5	0.129	0.278	0.160	0.295
image + metadata	10	0.074	0.310	0.111	0.239
image only	1	0.212	0.112	0.135	0.341
image only	5	0.086	0.220	0.112	0.198
image only	10	0.053	0.254	0.081	0.146

Table 3. Run 2 ablation, image + metadata.

ranking	F1@5	R@10
fusion (no hierarchy)	**0.172**	**0.324**
fusion + hierarchy	0.160	0.310
k-NN only	0.114	0.234
metadata lexical only	0.098	0.172
supervised head only	0.092	0.213
metadata text only	0.039	0.117
zero-shot only	0.016	0.076

Table 4. Run 2 recall@5 by category frequency in the index.

bucket	categories	labels	macro R
unseen (0)	508	1,309	0.009
tail (1–4)	44	84	0.333
torso (5–19)	23	213	0.457
head (20+)	47	222	0.758

Table 5. Run 2 by category type (k = 5).

type	labels	recall	precision
object	1,298	0.182	0.145
scene	338	0.210	0.149
event	55	0.273	0.080
abstract	137	0.051	0.028

Table 6. Run 2 robustness, F1@5 (share flagged low-quality).

degradation	flagged	+ metadata	image only
none	–	0.160	0.112
Gaussian blur r=2	1.00	0.152	0.099
Gaussian blur r=5	1.00	0.135	0.065
JPEG quality 10	0.22	0.156	0.095
downscale ×0.15	1.00	0.148	0.080

Efficiency (team's Mac, no server GPU): end-to-end per image p50 18.7 ms, p95 63.9 ms; scoring and ranking 0.73 ms/image; index of 2,906 images and 11,821 categories uses 28.8 MB in memory. By original resolution, F1@5 is 0.168 (≥ 2000 px), 0.133 (800–2000 px) and 0.166 (< 800 px).

Findings. (1) **More data helps across the board:** 2.7× more files raised F1@5 by 34% and P@1 from 0.284 to 0.339; mAP fell because the test set now spans 622 categories instead of 288, most of them new. (2) **Examples per category drive accuracy:** recall@5 is 0.758 for categories with ≥ 20 examples but 0.009 for unseen ones, and 72% of test labels belong to categories unseen in the index (62% are absent from the vocabulary entirely). This long tail of new categories is the central difficulty. (3) **Fusion works:** +51% F1@5 over the best single scorer (k-NN). (4) **Metadata matters:** +43% F1@5 over image only, and it halves the loss under heavy blur (−16% vs −42%). (5) **Strict metrics under-count:** hierarchy-aware P@1 is 0.482 vs strict 0.339. (6) **Weak spots:** zero-shot name matching alone is poor over 11.8k category names, and abstract concepts have the lowest recall.

Diagnosed issue. Validation F1@5 (0.307) is far above test (0.172) because the vocabulary is built from training *and* validation labels, so every validation label is "known", while 62% of test labels are not. Fusion weights and the hierarchy penalty are therefore tuned on an unrealistically easy set; this also explains why hierarchy suppression still costs 0.012 F1@5 on test. Fix: build the vocabulary only from data that precedes the validation period, fit fusion, then add validation categories (Run 3, below).

3. Progress on testing

- Evaluation harness complete with every metric planned in M1: P/R/F1@k, mAP, hierarchy-aware precision, frequency buckets, category types, resolution bands, synthetic degradations, latency and memory, plus per-scorer ablation.
- 60 automated tests: category filtering, API edge cases, splits, metrics, index search (FAISS and NumPy), hierarchy rules, the head, fusion, an offline end-to-end run (train → evaluate → predict → update) on synthetic data, and encoder API checks on tiny CLIP and SigLIP models.
- Continual learning verified: categories added at run time and accepted feedback are suggested immediately without retraining; rejected labels act as negative votes.

4. Plan for the remaining time

- **Run 3:** realistic validation vocabulary (above), then re-fit fusion and the hierarchy penalty; retrain only, no re-crawl.
- **Raise label coverage:** add category names without images (subtrees of the seeds, siblings of known categories) for the zero-shot and lexical scorers; track coverage and unseen-bucket recall.
- **Encoder comparison:** SigLIP-base vs CLIP ViT-B/32 on the same split, to strengthen zero-shot matching; report the accuracy/latency trade-off.
- **Reviewed score:** manually check a sample of "wrong" suggestions to estimate how many are correct but unlabelled.
- **Deliverables:** demo video (web UI with evidence and live feedback), final report, slides, externally hosted index.